DAVID E. J. VAN WIJK, PH.D.

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CURRENT POSITION

California Institute of Technology (August 2025 – Current)
Postdoctoral Scholar, Mechanical & Civil Engineering
Advisor: Prof. Joel Burdick

EDUCATION

Texas A&M University (2021 – August 2025)
Ph.D., Aerospace Engineering
Advisor: Prof. Manoranjan Majji
Dissertation: "Safety-Critical Control of Input-Constrained Systems"

Cornell University (2017 – 2021)
B.S., Mechanical & Aerospace Engineering *Magna Cum Laude*
Advisor: Prof. Silvia Ferrari

RESEARCH FOCUS AND SKILLS

I am a postdoctoral scholar passionate about developing **provably safe control algorithms for autonomous systems**. I am interested in problems which merge control theory, optimization, and estimation to enhance the reliability of autonomous operations in complex environments.

Research Interests: Safe autonomy, Control theory, Autonomous vehicles, Nonlinear systems, State estimation
Programming Languages: Python, MATLAB, C++, Git, LaTeX
Tools: Simscape Multibody, ANSYS, SolidWorks

PUBLICATIONS

* *indicates equal contribution*

UNDER REVIEW

R3. **D. E. J. van Wijk**, T. G. Molnar, S. Coogan, M. Majji, A. D. Ames, J. W. Burdick, "Output Feedback Backup Control Barrier Functions: Safety Guarantees Under Input Bounds and State Estimation Error," *Submitted to Systems & Control Letters*, 2026. (link)

R2. P. Ong, **D. E. J. van Wijk**, M. de Sa, J. W. Burdick, A. D. Ames, "SafeSpace: Aggregating Safe Sets from Backup Control Barrier Functions under Input Constraints," *Submitted to the IEEE Conference on Decision and Control (CDC)*, 2026. (link)

R1. **D. E. J. van Wijk**, E. Daş, A. Alan, S. Coogan, T. G. Molnar, J. W. Burdick, M. Majji, K. L. Hobbs, "Uncertainty Estimators for Robust Backup Control Barrier Functions," *Submitted to Automatica*, 2025. (link)

JOURNAL PUBLICATIONS

J7. **D. E. J. van Wijk**, D. Lee, E. Daş, T. G. Molnar, A. D. Ames, J. W. Burdick, "Generalizations of Backup Control Barrier Functions: Expansion and Adaptation for Input-Bounded Safety-Critical Control," *IEEE Control Systems Letters (L-CSS)*, 2026. (link)

J6. K. Echigo, **D. E. J. van Wijk**, P. Mestres, E. Daş, J. W. Burdick, A. D. Ames, "Probabilistic Control Barrier Functions for Systems with State Estimation Uncertainty using Sub-Gaussian Concentration," *IEEE Control Systems Letters (L-CSS)*, 2026. (to appear) (link)

J5. J. McElreath, **D. E. J. van Wijk**, M. Majji, "Controlling the Kalman Update: A Covariance Constrained Approach," *IEEE Transactions on Aerospace and Electronic Systems (TAES)*, 2025. (link)

J4. **D. E. J. van Wijk**, S. Coogan, T. G. Molnar, M. Majji, and K. L. Hobbs, "Disturbance-Robust Backup Control Barrier Functions: Safety Under Uncertain Dynamics," *IEEE Control Systems Letters (L-CSS)*, 2024. (link)

J3. I. Down, **D. E. J. van Wijk**, D. Parikh, M. Majji, "Autonomous Satellite Servicing Infrastructure for In-Space Assembly and Manufacturing," *ASME Journal of Manufacturing Science and Engineering, Special Issue on In-Space Manufacturing*, 2024. (link)

J2. **D. E. J. van Wijk**, K. Dunlap, M. Majji, and K. L. Hobbs, "Safe Spacecraft Inspection via Deep Reinforcement Learning and Discrete Control Barrier Functions," *AIAA Journal of Aerospace Information Systems (JAIS)*, 2024. (link)

J1. J. Gemerek, B. Fu, Y. Chen, Z. Liu, M. Zheng, **D. E. J. van Wijk**, S. Ferrari, "Directional Sensor Planning for Occlusion Avoidance," *IEEE Transactions on Robotics (T-RO)*, 2022. (link)

CONFERENCE PUBLICATIONS

C9. **D. E. J. van Wijk**, E. Daş, T. G. Molnar, A. D. Ames, J. W. Burdick, "Safety-Critical Control with Bounded Inputs: A Closed-Form Solution for Backup Control Barrier Functions," Proceedings of the *American Control Conference (ACC)*, New Orleans, Louisiana, USA, 26-29 May, 2026. (link)

C8. **D. E. J. van Wijk**, S. Coogan, T. G. Molnar, M. Majji, and K. L. Hobbs, "Disturbance-Robust Backup Control Barrier Functions: Safety Under Uncertain Dynamics," Proceedings of the *American Control Conference (ACC)*, Denver, Colorado, USA, 8-10 July 2025. (link)

C7. D. Parikh*, **D. E. J. van Wijk***, M. Majji, "Safe Multi-agent Satellite Servicing with Control Barrier Functions," Proceedings of the *Rocky Mountain AAS GN&C Conference*, Breckenridge, Colorado, USA, 1-5 February 2025. (link)

C6. K. Dunlap, K. Bennett, **D. E. J. van Wijk**, N. Hamilton, K. L. Hobbs, "Run Time Assured Reinforcement Learning for Six Degree-of-Freedom Spacecraft Inspection," Proceedings of the *AIAA ASCEND Conference*, July 2024. (link)

C5. **D. E. J. van Wijk***, I. Down*, and M. Majji, "On-Manifold Collision Avoidance using Tori Parametrization and Control Barrier Functions," Proceedings of the *Rocky Mountain AAS GN&C Conference*, Breckenridge, Colorado, USA, 1-7 February 2024.

C4. **D. E. J. van Wijk**, M. Majji, and K. L. Hobbs, "Fault Tolerant Run Time Assurance with Control Barrier Functions for Rigid Body Spacecraft Rotation," Proceedings of the *AIAA SciTech Forum*, Orlando, Florida, USA, 7-12 January 2024. (link)

C3. **D. E. J. van Wijk**, K. Dunlap, M. Majji, and K. L. Hobbs, "Deep Reinforcement Learning for Autonomous Spacecraft Inspection using Illumination," Proceedings of the *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana, USA, 13-17 August 2023. (link)

C2. K. Dunlap, **D. E. J. van Wijk**, and K. L. Hobbs, "Run Time Assurance for Autonomous Spacecraft Inspection," Proceedings of the *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana, USA, 13-17 August 2023. (link)

C1. **D. E. J. van Wijk**, K. Eves, and J. Valasek, "Deep Reinforcement Learning Controller for Autonomous Tracking of Evasive Ground Target," Proceedings of the *AIAA SciTech Forum*, National Harbor, Maryland, USA, 23-27 January 2023. (link)

OTHER PUBLICATIONS

O1. **D. E. J. van Wijk**, "Stochastic Control Barrier Functions for Economics," *arXiv preprint*, 2023. (link)

AWARDS AND HONORS

- **Texas A&M Graduate Excellence Fellowship Award (Fall 2022, Fall 2023)**: Open and highly competitive fellowship award of $1000, selected by the Texas A&M Department of Aerospace Engineering Graduate Program Committee.
- **Texas A&M Graduate Merit Fellowship (2021 – 2025)**: Highly competitive fellowship awarded to a single student in the Aerospace Engineering department per year.
- **Cornell University Dean's List**: Spring 2018; Fall 2018; Spring 2019; Fall 2019; Fall 2020; Spring 2021

OUTREACH AND MENTORSHIP

- **Student Mentorship**:
  - Dohyun Lee, Ph.D. Student, California Institute of Technology
  - Haru Tidmore, Ph.D. Student, Texas A&M University
- **Camp SOAR Outreach** (2022, 2023): Led laboratory tours and demonstrations for high school students interested in STEM.
- **Texas A&M Physics Festival** (2022, 2023, 2024): Led demonstrations for students K-12 to inspire the next generation of STEM students.

PROFESSIONAL SERVICE

- **Session Chair**: *AAS/AIAA Astrodynamics Specialist Conference 2023*, led and organized four technical sessions.
- **Reviewer**: *IEEE Control Systems Letters* (3), *IEEE Transactions on Automatic Control* (1), *IEEE International Conference on Robotics and Automation* (1), *IEEE Conference on Decision and Control* (1), *American Control Conference* (2),

AIAA Journal of Guidance, Control, and Dynamics (1), *AIAA Journal of Aerospace Information Systems* (3), *Journal of the Astronautical Sciences* (1), *AIAA SciTech Conference* (5), *AIAA Ascend* (3).

INVITED TALKS

I3. "Backup Control Barrier Functions for Run-Time Assurance." *2026 American Control Conference, Tutorial Session, New Orleans, LA*. May 2026.

I2. "Safety-Critical Control with Bounded Inputs." *46th Southern California Control Workshop, UC Irvine*. October 2025.

I1. "Fault Tolerant Run Time Assurance with Control Barrier Functions for Rigid Body Spacecraft Rotation." *Air Force Research Laboratory, Safe Trusted Autonomy for Responsible Spacecraft Annual Review*. October 2023.

SELECT RESEARCH EXPERIENCE

Safe Autonomy for Spacecraft Control (Aug '22 – Aug '25)
Research Internship with Dr. Kerianne Hobbs, Air Force Research Laboratory (AFRL), Safe Autonomy Team
Trained reinforcement learning agents for autonomous rendezvous, proximity operations, and docking (ARPOD) scenarios, and developed control barrier function based run-time assurance algorithms to guarantee safety of those agents. (J4, J2, C6, C4, C3, C2)

Simulation and Control of 7-DOF Spacecraft Manipulator (Aug '23 – Jan '24)
Graduate Research with Arkisys and Prof. Manoranjan Majji, Texas A&M University
Developed and validated a control system and manipulator planning system for 7-DOF robotic arm mounted on free-flying spacecraft using MATLAB Simulink and Simscape Multibody.

Spacecraft Manuever Classification using ML (Jan '23 – May '23)
Graduate Research with Ten One Aerospace, LLC and Prof. Manoranjan Majji, Texas A&M University
Built simulation of ground sensor and spacecraft dynamics to perform machine learning (ML) analysis for learning maneuver intent from ground sensor image traces.

Path Planning for Autonomous Drone (Jun '19 – May '21)
Undergraduate Research with Prof. Silvia Ferrari, Cornell University
Implemented and flight-tested custom algorithms for path planning in the presence of obstacles for an autonomous drone in occluded environments. (J1)

REFERENCES

Dr. Joel W. Burdick, Professor of Mechanical Engineering and Bioengineering; Jet Propulsion Laboratory Research Scientist, California Institute of Technology — contact: `jburdick@caltech.edu`

Dr. Aaron D. Ames, Professor of Mechanical and Civil Engineering, Control and Dynamical Systems, and Aerospace, California Institute of Technology — contact: `ames@caltech.edu`

Dr. Manoranjan Majji, Professor of Aerospace Engineering, Texas A&M University — contact: `mmajji@tamu.edu`

Dr. Kerianne L. Hobbs, Senior Engineering Specialist, Vehicle Autonomy & System Trust, The Aerospace Corporation — contact: `kerianne.hobbs@aero.org`

Dr. Dylan A. Shell, Professor of Computer Science and Engineering, Texas A&M University — contact: `dshell@tamu.edu`